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Department of CSE

B.Tech. CSE Trimester - 5

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Cloud Infrastructure and Services

25CS2206E

LAB RECORD

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CDIS – Y25

Experiment - 2

AIM: To enable Multi-Factor Authentication (MFA) on an AWS account to provide an additional layer of security, and to compare virtual MFA devices, hardware MFA devices and FIDO security keys.

Description:

Multi-Factor Authentication (MFA) adds an extra verification step when a user signs in to AWS. Instead of relying only on a password, the user must also prove possession of an approved authentication device or security key. This helps protect AWS resources if a password is lost, stolen or disclosed.

Steps:

Part A: Enable MFA for the AWS root user

1. Sign in to the AWS Management Console using the AWS account root user email address and password.
2. Open the account menu in the top navigation bar and choose **Security credentials** (or open the root user's security credentials page).
3. Find the **Multi-factor authentication (MFA)** section and choose **Assign MFA device**.
4. Enter a recognizable name for the MFA device.
5. Choose the MFA device type: virtual authenticator app, hardware TOTP token, or passkey/security key, depending on the available options.
6. For a virtual authenticator app, open an authenticator app on a phone or other device and scan the QR code displayed by AWS. If scanning is unavailable, use the setup key provided by AWS.
7. Enter the required consecutive authentication codes from the authenticator app and complete the assignment process. For a passkey/security key, follow the on-screen prompts to register the device.

8. After successful registration, confirm that the MFA device appears in the MFA section of the security credentials page.

9. Sign out and test the sign-in process, confirming that AWS requests the additional MFA verification when required.

Part B: Comparison of MFA device types

Virtual MFA device	A smartphone or other device runs an authentic generates time-based one-time passwords (TOTP). Co available; requires access to	
Hardware MFA device	A dedicated physical token generates authenticat physical device; must be kept safe and ava	
FIDO security key	A physical security key or supported passkey uses pub for authentication. Strong phishing-resistant authe	

Output:

MFA provides an additional authentication factor for AWS sign-in. After successful setup, the configured MFA method is listed in the account or user's security credentials page, and AWS can request an MFA verification during sign-in.

Screenshot evidence note: The supplied screenshot shows the IAM user named **DHARSHAN** with one registered device under Multi-factor authentication. Its type is **Passkeys and security keys**. This is evidence of MFA registration for that IAM user; the screenshot does not establish that MFA was enabled on the AWS root user.

MFA Configuration – Screenshot Evidence

The screenshot below was supplied as evidence of an MFA device registered in the AWS IAM security credentials page.

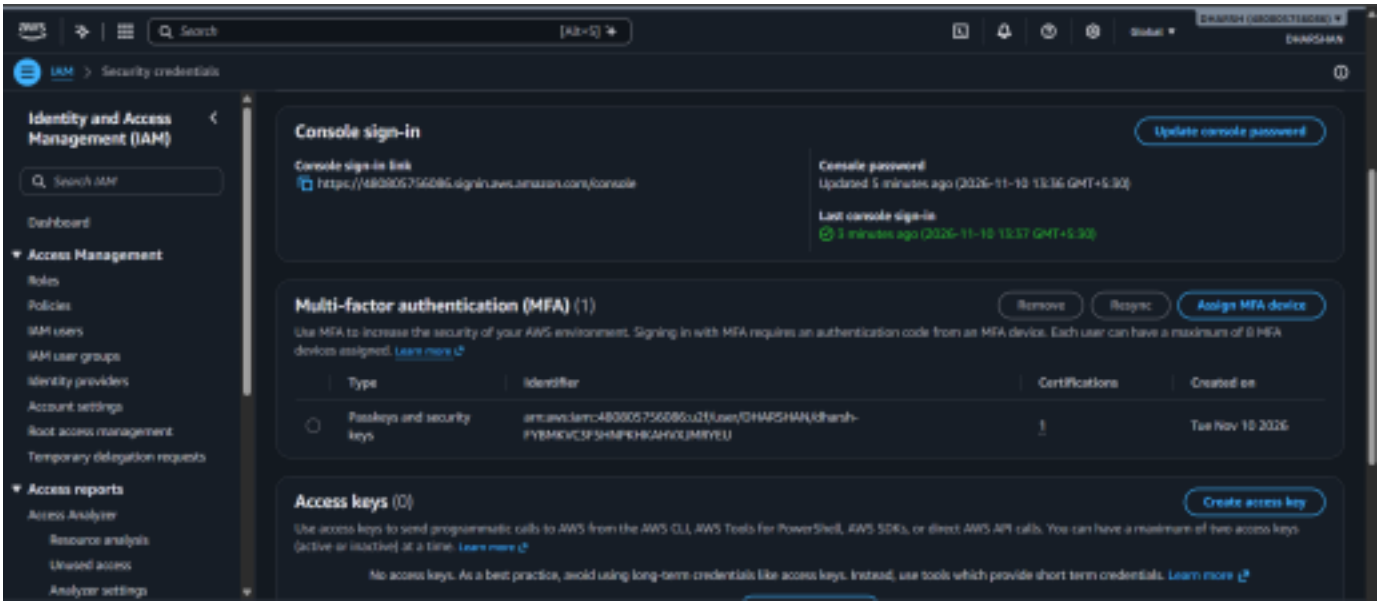


Figure: AWS IAM Security credentials page showing one registered passkey/security key MFA device.